

[FAMA AI]

FAMA AI - FAMA-FRENCH 3-FACTOR STOCK ANALYSIS PLATFORM

Comprehensive Program Source Code Reference Catalog

Author / Project: MBA + BTech Academic Team Research Project

Technology Stack: FastAPI (Python), Statsmodels OLS, React.js, Tailwind CSS, Plotly.js

Document Scope: 100% Unabridged Source Code Library

System Core: Ordinary Least Squares Factor Pricing Engine

■ backend/main.py

Component Description: FastAPI Server application endpoint routing, static asset mounting, CORS configs, and dynamic PDF file streaming handlers.

```
import os
from fastapi import FastAPI, UploadFile, File, HTTPException, Form
from fastapi.middleware.cors import CORSMiddleware
from fastapi.responses import StreamingResponse, FileResponse
from fastapi.staticfiles import StaticFiles
from pydantic import BaseModel
from typing import Dict, Any, List

from backend.utils.csv_loader import load_csv
from backend.services.regression_service import run_fama_french_regression
from backend.utils.pdf_generator import generate_pdf_report

app = FastAPI(
    title="FAMA AI - Fama-French 3-Factor Stock Analysis Platform",
    description="Backend API and web server for statistical equity factor analysis."
)

# Enable CORS for local development
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)

# Sample stock mapping
SAMPLE_STOCKS = {
    "infosys": {
        "name": "Infosys Limited (INFY)",
        "file_name": "infosys.csv"
    },
    "tata_motors": {
        "name": "Tata Motors Limited (TATAMOTORS)",
        "file_name": "tata_motors.csv"
    }
}

class ReportRequest(BaseModel):
    stock_name: str
    results: Dict[str, Any]

@app.get("/api/stocks")
def get_stocks():
    """
    Returns the list of pre-configured sample stocks.
    """
    return [
        {"id": symbol, "name": meta["name"]}
        for symbol, meta in SAMPLE_STOCKS.items()
    ]

@app.get("/api/fama-analysis/{symbol}")
def analyze_sample_stock(symbol: str):
    """
    Loads a sample stock factor dataset, performs regression, and returns raw data + results.
    """
    if symbol not in SAMPLE_STOCKS:
        raise HTTPException(
```

```

        status_code=404,
        detail=f"Stock symbol {symbol} not found. Choose {infosys} or {tata_motors}"
    )

stock_meta = SAMPLE_STOCKS[symbol]
file_path = os.path.join("backend", "sample_data", stock_meta["file_name"])

if not os.path.exists(file_path):
    raise HTTPException(
        status_code=500,
        detail=f"Sample data file {stock_meta["file_name"]} is missing on the server."
    )

try:
    # Load and validate
    df = load_csv(file_path)
    # Run regression
    results = run_fama_french_regression(df)
    # Prepare raw data points for graphs
    raw_data = df.to_dict(orient="records")

    return {
        "stock_name": stock_meta["name"],
        "results": results,
        "data": raw_data
    }
except Exception as e:
    raise HTTPException(status_code=400, detail=str(e))

@app.post("/api/fama-analysis/upload")
async def analyze_uploaded_csv(file: UploadFile = File(...)):
    """"
    Validates uploaded CSV file, runs Fama-French regression, and returns raw data + results.
    """"
    if not file.filename.endswith(".csv"):
        raise HTTPException(status_code=400, detail="Only CSV files are accepted.")

    try:
        contents = await file.read()
        # Load and validate CSV from bytes
        df = load_csv(contents)
        # Run regression
        results = run_fama_french_regression(df)
        # Raw records
        raw_data = df.to_dict(orient="records")

        # Extrapolate name from filename
        stock_name = os.path.splitext(file.filename)[0].replace("_", " ").title() + " (Uploaded)"

        return {
            "stock_name": stock_name,
            "results": results,
            "data": raw_data
        }
    except Exception as e:
        raise HTTPException(status_code=400, detail=str(e))

@app.post("/api/reports/export")
def export_pdf_report(payload: ReportRequest):
    """"
    Stateless endpoint that generates and downloads the OLS report PDF based on client inputs.
    """"
    try:
        pdf_buffer = generate_pdf_report(payload.stock_name, payload.results)

        # Clean filename

```

```
safe_name = "".join(c for c in payload.stock_name if c.isalnum() or c in ('&#x27; &#x27;; &#x27;-&#x27;', &
safe_name = safe_name.replace('" &quot;; &quot;_&quot;')
filename = f"Fama_French_Report_{safe_name}.pdf"

return StreamingResponse(
    pdf_buffer,
    media_type="application/pdf",
    headers={"Content-Disposition": f"attachment; filename={filename}"})
)
except Exception as e:
    raise HTTPException(status_code=500, detail=f"PDF generation failed: {str(e)}")

# Fallback serving of SPA root index.html
@app.get("/&quot;)
def serve_index():
    index_path = os.path.join("backend", "static", "index.html")
    if os.path.exists(index_path):
        return FileResponse(index_path)
    return {"message": "FAMA AI server is running. Frontend static directory is empty or setup failed."}

# Mount static files (javascript, styles, favicon)
# Must be mounted AFTER other specific endpoints to avoid route interception.
static_dir = os.path.join("backend", "static")
if os.path.exists(static_dir):
    app.mount("/&quot;, StaticFiles(directory=static_dir, html=True), name="static")
else:
    # Create directory structure for static files
    os.makedirs(os.path.join(static_dir, "js"), exist_ok=True)
    os.makedirs(os.path.join(static_dir, "css"), exist_ok=True)
    app.mount("/&quot;, StaticFiles(directory=static_dir, html=True), name="static")
```

■ backend/services/regression_service.py

Component Description: OLS Factor Regression executor utilizing statsmodels. Computes coefficients, standard errors, p-values, t-stats, R-squared, and renders plain-English explanations.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from typing import Dict, Any

def run_fama_french_regression(df: pd.DataFrame) -> Dict[str, Any]:
    """
    Performs Fama-French 3-Factor OLS regression.
    Formula:  $R_{i,Rf} = \alpha + \beta_1(R_{m,Rf}) + \beta_2(SMB) + \beta_3(HML)$ 

    Parameters:
        df: pd.DataFrame with Date, Ri_Rf, Rm_Rf, SMB, HML

    Returns:
        Dict containing OLS summary statistics, coefficients, p-values, t-stats,
        R-squared, confidence intervals, and qualitative interpretations.
    """
    # Define variables
    y = df[["Ri_Rf"]]
    X = df[["Rm_Rf", "SMB", "HML"]]

    # Add constant for Alpha (intercept)
    X_with_const = sm.add_constant(X)

    # Run OLS regression
    model = sm.OLS(y, X_with_const)
    results = model.fit()

    # Extract coefficients
    params = results.params
    pvalues = results.pvalues
    tvalues = results.tvalues
    conf_int = results.conf_int()
    bse = results.bse # Standard errors

    # Output structure
    regression_stats = {
        "num_observations": int(results.nobs),
        "r_squared": float(results.rsquared),
        "adj_r_squared": float(results.rsquared_adj),
        "f_statistic": float(results.fvalue),
        "f_pvalue": float(results.f_pvalue),
        "factors": {
            "Alpha": {
                "coefficient": float(params["const"]),
                "p_value": float(pvalues["const"]),
                "t_stat": float(tvalues["const"]),
                "std_err": float(bse["const"]),
                "conf_lower": float(conf_int.loc["const", 0]),
                "conf_upper": float(conf_int.loc["const", 1]),
                "significant": bool(pvalues["const"] < 0.05)
            },
            "Market": {
                "coefficient": float(params["Rm_Rf"]),
                "p_value": float(pvalues["Rm_Rf"]),
                "t_stat": float(tvalues["Rm_Rf"]),
                "std_err": float(bse["Rm_Rf"]),
                "conf_lower": float(conf_int.loc["Rm_Rf", 0]),
                "conf_upper": float(conf_int.loc["Rm_Rf", 1]),
            }
        }
    }
```

```

        &quot;significant&quot;: bool(pvalues[&#x27;Rm_Rf&#x27;] &lt; 0.05)
    },
    &quot;SMB&quot;: {
        &quot;coefficient&quot;: float(params[&#x27;SMB&#x27;]),
        &quot;p_value&quot;: float(pvalues[&#x27;SMB&#x27;]),
        &quot;t_stat&quot;: float(tvalues[&#x27;SMB&#x27;]),
        &quot;std_err&quot;: float(bse[&#x27;SMB&#x27;]),
        &quot;conf_lower&quot;: float(conf_int.loc[&#x27;SMB&#x27;, 0]),
        &quot;conf_upper&quot;: float(conf_int.loc[&#x27;SMB&#x27;, 1]),
        &quot;significant&quot;: bool(pvalues[&#x27;SMB&#x27;] &lt; 0.05)
    },
    &quot;HML&quot;: {
        &quot;coefficient&quot;: float(params[&#x27;HML&#x27;]),
        &quot;p_value&quot;: float(pvalues[&#x27;HML&#x27;]),
        &quot;t_stat&quot;: float(tvalues[&#x27;HML&#x27;]),
        &quot;std_err&quot;: float(bse[&#x27;HML&#x27;]),
        &quot;conf_lower&quot;: float(conf_int.loc[&#x27;HML&#x27;, 0]),
        &quot;conf_upper&quot;: float(conf_int.loc[&#x27;HML&#x27;, 1]),
        &quot;significant&quot;: bool(pvalues[&#x27;HML&#x27;] &lt; 0.05)
    }
}

# Generate qualitative interpretations
regression_stats[&quot;interpretations&quot;] = generate_interpretations(regression_stats)

return regression_stats

def generate_interpretations(stats: Dict[str, Any]) -> Dict[str, str]:
    &quot;&quot;&quot;
    Generates plain-English financial interpretations of regression metrics.
    &quot;&quot;&quot;
    factors = stats[&quot;factors&quot;]
    r2 = stats[&quot;r_squared&quot;]

    mkt = factors[&quot;Market&quot;]
    smb = factors[&quot;SMB&quot;]
    hml = factors[&quot;HML&quot;]
    alpha = factors[&quot;Alpha&quot;]

    interpretations = {}

    # 1. Market Beta Interpretation
    mkt_coeff = mkt[&quot;coefficient&quot;]
    mkt_sig = &quot; (statistically significant)&quot; if mkt[&quot;significant&quot;] else &quot; (not statistically significant)&quot;
    if mkt_coeff > 1.1:
        interpretations[&quot;market&quot;] = f&quot;This stock is highly sensitive to market movements (Beta = {mkt_coeff:.2f}). Its performance is {mkt_sig}&quot;
    elif mkt_coeff < 0.9:
        interpretations[&quot;market&quot;] = f&quot;This stock is relatively stable compared to the market (Beta = {mkt_coeff:.2f}). Its performance is {mkt_sig}&quot;
    else:
        interpretations[&quot;market&quot;] = f&quot;This stock moves in sync with the broad market index (Beta = {mkt_coeff:.2f}). Its performance is {mkt_sig}&quot;

    # 2. SMB (Size Factor) Interpretation
    smb_coeff = smb[&quot;coefficient&quot;]
    smb_sig = &quot; (statistically significant)&quot; if smb[&quot;significant&quot;] else &quot; (not statistically significant)&quot;
    if smb_coeff > 0.15:
        interpretations[&quot;smb&quot;] = f&quot;The stock shows characteristics associated with smaller companies (SMB Beta = {smb_coeff:.2f}). Its performance is {smb_sig}&quot;
    elif smb_coeff < -0.15:
        interpretations[&quot;smb&quot;] = f&quot;The stock behaves more like large-cap companies (SMB Beta = {smb_coeff:.2f}). Its performance is {smb_sig}&quot;
    else:
        interpretations[&quot;smb&quot;] = f&quot;The stock is size-neutral (SMB Beta = {smb_coeff:.2f}). Its performance is {smb_sig}&quot;

    # 3. HML (Value vs Growth Factor) Interpretation
    hml_coeff = hml[&quot;coefficient&quot;]
    hml_sig = &quot; (statistically significant)&quot; if hml[&quot;significant&quot;] else &quot; (not statistically significant)&quot;
    if hml_coeff > 0.15:
        interpretations[&quot;hml&quot;] = f&quot;The stock shows characteristics associated with value companies (HML Beta = {hml_coeff:.2f}). Its performance is {hml_sig}&quot;
    elif hml_coeff < -0.15:
        interpretations[&quot;hml&quot;] = f&quot;The stock shows characteristics associated with growth companies (HML Beta = {hml_coeff:.2f}). Its performance is {hml_sig}&quot;
    else:
        interpretations[&quot;hml&quot;] = f&quot;The stock is market-neutral (HML Beta = {hml_coeff:.2f}). Its performance is {hml_sig}&quot;

    return interpretations

```

```

        interpretations["hml"] = f"The stock exhibits value-oriented characteristics (HML Beta = {hml_coeff:.3f})"
    elif hml_coeff < -0.15:
        interpretations["hml"] = f"The stock exhibits growth-oriented characteristics (HML Beta = {hml_coeff:.3f})"
    else:
        interpretations["hml"] = f"The stock behaves like a blend of value and growth (HML Beta = {hml_coeff:.3f})"

# 4. Alpha (Abnormal Return) Interpretation
alpha_coeff = alpha["coefficient"] * 100 # Convert to percent
alpha_sig = " (statistically significant)" if alpha["significant"] else " (not statistically significant)"
if alpha["significant"] and alpha_coeff > 0:
    interpretations["alpha"] = f"The stock generated a positive abnormal return of {alpha_coeff:.3f}% per period (Alpha = {alpha_coeff:.3f}%)"
elif alpha["significant"] and alpha_coeff < 0:
    interpretations["alpha"] = f"The stock underperformed by {alpha_coeff:.3f}% per period (Alpha = {alpha_coeff:.3f}%)"
else:
    interpretations["alpha"] = f"The stock's abnormal return (Alpha = {alpha['coefficient']:.3f})"

# 5. R-squared Interpretation
r2_pct = r2 * 100
if r2 > 0.85:
    interpretations["r_squared"] = f"The Fama-French 3-Factor model explains a very large portion ({r2_pct:.1f})% of the stock's return."
elif r2 >= 0.50:
    interpretations["r_squared"] = f"The Fama-French 3-Factor model explains a moderate portion ({r2_pct:.1f})% of the stock's return."
else:
    interpretations["r_squared"] = f"The Fama-French 3-Factor model explains a small portion ({r2_pct:.1f})% of the stock's return."

return interpretations

```

▣ backend/utils/csv_loader.py

Component Description: Robust CSV parsing engine. Standardizes columns, handles whitespaces, validates file data format, and raises clean error messages.

```
import pandas as pd
import io
from typing import Union
```

```
REQUIRED_COLUMNS = ["Date", "Ri_Rf", "Rm_Rf", "SMB", "HML"]
```

```
def load_csv(file_source: Union[str, bytes, io.BytesIO]) -> pd.DataFrame:
    """
    Loads and validates a Fama-French CSV dataset.
```

Parameters:

file_source: Can be a file path (str), raw bytes (from file upload), or a BytesIO stream.

Returns:

pd.DataFrame containing columns: Date, Ri_Rf, Rm_Rf, SMB, HML

Raises:

ValueError: If required columns are missing, data is empty, or numeric columns contain non-numeric data.

try:

```
    if isinstance(file_source, bytes):
        df = pd.read_csv(io.BytesIO(file_source))
    elif isinstance(file_source, io.BytesIO):
        df = pd.read_csv(file_source)
    else:
        # Assume it's a file path string
        df = pd.read_csv(file_source)
```

except Exception as e:

```
    raise ValueError(f"Failed to read CSV file: {str(e)}")
```

if df.empty:

```
    raise ValueError("The uploaded CSV file is empty.")
```

Clean up column names (strip whitespace and handle casing)

```
df.columns = [col.strip() for col in df.columns]
```

Case-insensitive column matching

```
column_mapping = {}
```

```
lower_required = [col.lower() for col in REQUIRED_COLUMNS]
```

for col in df.columns:

```
    if col.lower() in lower_required:
        # Map the actual column name to the standard capitalized name
        idx = lower_required.index(col.lower())
        column_mapping[col] = REQUIRED_COLUMNS[idx]
```

Check if all required columns are found

```
missing_cols = [col for col in REQUIRED_COLUMNS if col not in column_mapping.values()]
```

if missing_cols:

```
    raise ValueError(f"Missing required columns: {', '.join(missing_cols)}. The CSV must contain Date, R
```

Rename columns to our standard format

```
df = df.rename(columns=column_mapping)
```

Select only the required columns

```
df = df[REQUIRED_COLUMNS]
```

Drop rows where all elements are NaN

```
df = df.dropna(how='all')
```



```
# Check if we have at least 3 rows to run OLS regression
if len(df) < 4:
    raise ValueError(""At least 4 data points are required to run the Fama-French 3-Factor regression."")

# Validate Date column
df[Date] = df[Date].astype(str).str.strip()
if df[Date].isnull().any() or (df[Date] == "").any():
    raise ValueError(""The Date column contains empty or null values."")

# Validate numeric columns
numeric_cols = ["Ri_Rf", "Rm_Rf", "SMB", "HML"]
for col in numeric_cols:
    try:
        df[col] = pd.to_numeric(df[col])
    except Exception:
        raise ValueError(f""Column {col} must contain only numeric values."")

# Drop rows where numeric columns are NaN
df = df.dropna(subset=[col])

if len(df) < 4:
    raise ValueError(""Not enough valid numeric rows to perform regression analysis."")

return df
```

□ backend/utils/pdf_generator.py

Component Description: PDF Document generator utilizing ReportLab. Builds two-column layout statistical regression summaries in-memory.

```
import io
from reportlab.lib.pagesizes import letter
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib import colors
from typing import Dict, Any

def generate_pdf_report(stock_name: str, results: Dict[str, Any]) -> io.BytesIO:
    """
    Generates a structured, presentation-ready PDF report of the OLS Fama-French analysis.

    Parameters:
        stock_name: Name of the stock analyzed.
        results: Dict from run_fama_french_regression.

    Returns:
        BytesIO stream of the PDF file.
    """
    buffer = io.BytesIO()

    # Page setup
    doc = SimpleDocTemplate(
        buffer,
        pagesize=letter,
        rightMargin=40,
        leftMargin=40,
        topMargin=40,
        bottomMargin=40
    )

    styles = getSampleStyleSheet()

    # Custom styles
    title_style = ParagraphStyle(
        '<#x27;DocTitle<#x27;;',
        parent=styles['<#x27;Heading1<#x27;'],
        fontName='<#x27;Helvetica-Bold<#x27;',
        fontSize=24,
        textColor=colors.HexColor('<#x27;#0F172A<#x27;'), # Slate 900
        spaceAfter=12
    )

    subtitle_style = ParagraphStyle(
        '<#x27;DocSubtitle<#x27;;',
        parent=styles['<#x27;Normal<#x27;'],
        fontName='<#x27;Helvetica<#x27;',
        fontSize=11,
        textColor=colors.HexColor('<#x27;#64748B<#x27;'), # Slate 500
        spaceAfter=20
    )

    heading2_style = ParagraphStyle(
        '<#x27;SectionHeading<#x27;;',
        parent=styles['<#x27;Heading2<#x27;'],
        fontName='<#x27;Helvetica-Bold<#x27;',
        fontSize=15,
        textColor=colors.HexColor('<#x27;#1E3A8A<#x27;'), # Navy blue
        spaceBefore=14,
        spaceAfter=8,
```

```

        keepWithNext=True
    )

    body_style = ParagraphStyle(
        &#x27;BodyTextCustom&#x27;,
        parent=styles[&#x27;BodyText&#x27;],
        fontName=&#x27;Helvetica&#x27;,
        fontSize=10,
        textColor=colors.HexColor(&#x27;#334155&#x27;), # Slate 700
        leading=14
    )

    bullet_style = ParagraphStyle(
        &#x27;BulletCustom&#x27;,
        parent=styles[&#x27;Normal&#x27;],
        fontName=&#x27;Helvetica&#x27;,
        fontSize=9.5,
        textColor=colors.HexColor(&#x27;#1E293B&#x27;), # Slate 800
        leftIndent=15,
        firstLineIndent=-10,
        spaceAfter=6,
        leading=13
    )

    formula_style = ParagraphStyle(
        &#x27;FormulaText&#x27;,
        parent=styles[&#x27;Normal&#x27;],
        fontName=&#x27;Courier-Bold&#x27;,
        fontSize=10,
        textColor=colors.HexColor(&#x27;#312E81&#x27;), # Indigo 900
        alignment=1, # Centered
        backgroundColor=colors.HexColor(&#x27;#EEF2F6&#x27;),
        borderPadding=8,
        spaceAfter=15
    )

    elements = []

    # Header Section
    elements.append(Paragraph(&quot;FAMA AI&quot;;, title_style))
    elements.append(Paragraph(f&quot;Fama-French 3-Factor Regression Report &amp;bull; Academic Analysis &amp;bull; {stock_

# Divider line
divider = Table([[&quot;&quot;]], colWidths=[532], rowHeights=[2])
divider.setStyle(TableStyle([
    (&#x27;BACKGROUND&#x27;, (0,0), (-1,-1), colors.HexColor(&#x27;#1E3A8A&#x27;)),
    (&#x27;TOPPADDING&#x27;, (0,0), (-1,-1), 0),
    (&#x27;BOTTOPADDING&#x27;, (0,0), (-1,-1), 0),
]))
elements.append(divider)
elements.append(Spacer(1, 15))

# Executive Summary / Regression Equation
elements.append(Paragraph(&quot;Regression Equation&quot;;, heading2_style))
equation_text = (
    &quot;R<sup>i</sup> - R<sup>f</sup> = &amp;alpha; + &amp;beta;<sub>Market</sub>
    &quot;s &amp;times; SMB + h &amp;times; HML + &amp;epsilon;&quot;
)
elements.append(Paragraph(equation_text, formula_style))

# Stock Overview Description
intro_p = (
    f&quot;This academic report presents the OLS regression analysis results for &lt;b>{stock_name}&lt;/b> &quot;
    f&quot;using the Fama-French 3-Factor Model. The analysis calculates exposure to systemic risk factors &quot;
    f&quot;over {results[&#x27;num_observations&#x27;]} observation periods. Below are the coefficients, p-values, &quot;
    f&quot;statistical significance, and economic interpretations.&quot;

```

```

)
elements.append(Paragraph(intro_p, body_style))
elements.append(Spacer(1, 12))

# Regression Results Table
elements.append(Paragraph(""Model Coefficients & Statistics"", heading2_style))

factors = results["factors"]

# Prepare data for table
table_data = [
    [""Factor"", ""Coefficient"", ""Std Error"", ""t-Statistic"", ""p-Value""],
]

for factor_name, data in factors.items():
    # Map standard keys
    display_name = {
        ""Alpha"": ""Alpha (Intercept)"",
        ""Market"": ""Market Premium (Rm-Rf)"",
        ""SMB"": ""Size Factor (SMB)"",
        ""HML"": ""Style Factor (HML)""
    }.get(factor_name, factor_name)

    conf_interval = f""[{data['conf_lower']:.4f}, {data['conf_upper']:.4f}]""
    sig_text = ""Yes (p < 0.05)"" if data["significant"] else ""No""

    table_data.append([
        display_name,
        f""{data['coefficient']:.4f}"",
        f""{data['std_err']:.4f}"",
        f""{data['t_stat']:.2f}"",
        f""{data['p_value']:.4f}"",
        conf_interval,
        sig_text
    ])

# Table layout & styles
col_widths = [135, 65, 55, 60, 50, 110, 57]
stats_table = Table(table_data, colWidths=col_widths)
stats_table.setStyle(TableStyle([
    (&#x27;BACKGROUND&#x27;, (0,0), (-1,0), colors.HexColor(&#x27;#0F172A&#x27;)), # Dark navy header
    (&#x27;TEXTCOLOR&#x27;, (0,0), (-1,0), colors.whitesmoke),
    (&#x27;ALIGN&#x27;, (0,0), (-1,-1), &#x27;CENTER&#x27;),
    (&#x27;ALIGN&#x27;, (0,1), (0,-1), &#x27;LEFT&#x27;), # Left align first column
    (&#x27;FONTNAME&#x27;, (0,0), (-1,0), &#x27;Helvetica-Bold&#x27;),
    (&#x27;FONTSIZE&#x27;, (0,0), (-1,-1), 8),
    (&#x27;BOTTPADDING&#x27;, (0,0), (-1,-1), 6),
    (&#x27;TOPPADDING&#x27;, (0,0), (-1,-1), 6),
    (&#x27;GRID&#x27;, (0,0), (-1,-1), 0.5, colors.HexColor(&#x27;#CBD5E1&#x27;)), # Light grey border
    (&#x27;ROWBACKGROUNDS&#x27;, (0,1), (-1,-1), [colors.white, colors.HexColor(&#x27;#F8FAFC&#x27;)]) # Alternating r
]))
elements.append(stats_table)
elements.append(Spacer(1, 10))

# Model Diagnostics Table
elements.append(Paragraph(""Model Fit Diagnostics"", heading2_style))
diagnostics_data = [
    Paragraph("<b>Number of Observations:</b>", body_style), f""{results['num_observ
    Paragraph("<b>R-Squared (R&sup2;)</b>", body_style), f""{results['r_squared
],
    Paragraph("<b>F-Statistic:</b>", body_style), f""{results['f_statistic&#x27;]:.2
    Paragraph("<b>Adjusted R-Squared:</b>", body_style), f""{results['adj_r_squared&
],
    [

```

```

        Paragraph("<b>Prob (F-statistic):</b>", body_style), f"<results[f_pvalue]
        "</>", "</>"
    ]
]
diag_table = Table(diagnostics_data, colWidths=[150, 100, 150, 132])
diag_table.setStyle(TableStyle([
    (<VALIGN, (0,0), (-1,-1), MIDDLE),
    (<BOTTOMPADDING, (0,0), (-1,-1), 4),
    (<TOPPADDING, (0,0), (-1,-1), 4),
    (<LINEBELOW, (0,0), (-1,-1), 0.5, colors.HexColor("#E2E8F0")),
    (<SPAN, (2,2), (3,2)), # Span empty cells
]))
elements.append(diag_table)
elements.append(Spacer(1, 15))

# Qualitative Interpretations
elements.append(Paragraph("Fama-French Factor Interpretation", heading2_style))

interpretations = results["interpretations"]

bullets = [
    "<b>Market Sensitivity (&beta;):</b> {interpretations[market]}",
    "<b>Size Premium Exposure (SMB):</b> {interpretations[smb]}",
    "<b>Style Premium Exposure (HML):</b> {interpretations[hml]}",
    "<b>Abnormal Return (&alpha;):</b> {interpretations[alpha]}",
    "<b>Model Explanatory Power (R&sup2;):</b> {interpretations[r_squared]}",
]

for bullet in bullets:
    elements.append(Paragraph(f"&bull; {bullet}", bullet_style))

elements.append(Spacer(1, 20))

# Academic Disclaimer Footer
disclaimer_style = ParagraphStyle(
    "<Disclaimer>",
    parent=styles["Normal"],
    fontName="Helvetica-Oblique",
    fontSize=8,
    textColor=colors.HexColor("#94A3B8"),
    alignment=1, # Centered
    spaceBefore=20
)
elements.append(Paragraph(
    "<FAMA AI is an academic analysis project. This analysis is based on historical factor data and </>
    </>should not be construed as investment advice. Past performance is not indicative of future results.</>
    disclaimer_style
))

# Build Document
doc.build(elements)
buffer.seek(0)
return buffer

```

📁 backend/static/index.html

Component Description: Client SPA page skeleton. Loads CDN React, Tailwind CSS v3 configuration, Plotly.js library, and configures error diagnosis overlays.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>FAMA AI - Fama-French 3-Factor Stock Analysis Platform</title>

  <!-- Global Error Handler to Diagnose Startup Crashes -->
  <script>
    window.addEventListener("error", function(e) {
      var root = document.getElementById("root");
      if (root) {
        root.innerHTML = `
          <div class="min-h-screen flex flex-col items-center justify-center bg-slate-950 p-6 font-sans">
            <div class="max-w-xl w-full bg-rose-950/20 border border-rose-500/30 p-6 rounded-xl text-rose-200">
              <h3 class="text-lg font-bold mb-2 text-rose-400">🚨 Front-End Runtime Error</h3>
              <p class="text-sm font-semibold mb-1">Message: ${e.message}</p>
              <p class="text-xs font-mono text-gray-400">File: ${e.filename} (Line ${e.lineno}, Col ${e.colno})</p>
              <p class="text-xs text-gray-500 mt-4 leading-relaxed">Please send this error report back to A
            </div>
          </div>
        `;
      }
    });
  </script>

  <!-- SEO Meta Tags -->
  <meta name="description" content="An academic equity research tool that automates Fama-French 3-Factor
  <meta name="keywords" content="Fama-French, Stock Analysis, OLS Regression, NIFTY 50, Finance, MBA Proj

  <!-- Tailwind CSS CDN -->
  <script src="https://cdn.tailwindcss.com"></script>

  <!-- Tailwind Configuration for Slate overrides and custom styles -->
  <script>
    tailwind.config = {
      theme: {
        extend: {
          colors: {
            slate: {
              950: "#071212",
              900: "#0f172a",
              850: "#111827",
              800: "#1e293b",
              700: "#334155",
            }
          }
        }
      }
    }
  </script>

  <!-- Custom Stylesheet -->
  <link rel="stylesheet" href="/css/styles.css">

  <!-- React & React DOM CDNs -->
  <script crossorigin src="https://unpkg.com/react@18.2.0/umd/react.production.min.js"></script>
  <script crossorigin src="https://unpkg.com/react-dom@18.2.0/umd/react-dom.production.min.js"></script>
  <!-- Plotly.js CDN -->
```

```
<script src="https://cdn.plot.ly/plotly-2.24.1.min.js" charset="utf-8"></script>
</head>
<body class="bg-slate-950 text-gray-100 min-h-screen">

  <!-- React App Mount Point -->
  <div id="root">
    <!-- Server-Side Pre-loader for Premium First Impression -->
    <div class="min-h-screen flex flex-col items-center justify-center bg-slate-950 text-gray-100">
      <div class="flex flex-col items-center gap-4 animate-pulse">
        <div class="w-12 h-12 rounded-xl bg-gradient-to-tr from-indigo-500 to-cyan-400 flex items-center justify-center">
          <svg class="w-7 h-7 text-slate-900" fill="none" viewBox="0 0 24 24" stroke="none">
            <path stroke-linecap="round" stroke-linejoin="round" d="M12 6.253v13m0-13C10.832 5.
          </svg>
        </div>
        <div class="text-center">
          <h2 class="text-lg font-bold tracking-wider text-white uppercase font-sans">FAMA AI</h2>
          <p class="text-xs text-gray-500 font-mono mt-1">Bootstrapping Client Components...</p>
        </div>
      </div>
    </div>

    <!-- Main Modular Application Entrypoint -->
    <script type="module" src="/js/app.js"></script>
  </div>
</body>
</html>
```

📄 backend/static/css/styles.css

Component Description: Application UI stylesheets. Configures custom dark theme elements, scrollbars, glassmorphic cards, and title keyframe animations.

```
@import url(&#x27;https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&family=Outfit:wght@400;500;600);

/* HTML scroll behavior for smooth transition from hero to dashboard */
html {
  scroll-behavior: smooth;
}

/* Main Body Configuration */
body {
  font-family: &#x27;Inter&#x27;;, sans-serif;
  background-color: #030712; /* Slate-950 equivalent */
  color: #f3f4f6; /* Gray-100 equivalent */
  overflow-x: hidden;
}

/* Headings font */
h1, h2, h3, h4, h5, h6 {
  font-family: &#x27;Outfit&#x27;;, sans-serif;
}

/* Monospace stats/math font */
.mono-font {
  font-family: &#x27;Fira Code&#x27;;, monospace;
}

/* Custom Sleek Scrollbar */
::-webkit-scrollbar {
  width: 6px;
  height: 6px;
}

::-webkit-scrollbar-track {
  background: #0b0f19;
}

::-webkit-scrollbar-thumb {
  background: #1f2937;
  border-radius: 3px;
}

::-webkit-scrollbar-thumb:hover {
  background: #374151;
}

/* Glassmorphism Styles */
.glass-panel {
  background: rgba(17, 24, 39, 0.7);
  backdrop-filter: blur(12px);
  -webkit-backdrop-filter: blur(12px);
  border: 1px solid rgba(255, 255, 255, 0.06);
  border-radius: 1rem;
}

.glass-panel-hover {
  transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
}

.glass-panel-hover:hover {
  transform: translateY(-2px);
}
```



```
background: rgba(17, 24, 39, 0.85);
border-color: rgba(99, 102, 241, 0.3) !important; /* Indigo glow on hover */
box-shadow: 0 10px 25px -5px rgba(0, 0, 0, 0.4),
            0 0 20px 0 rgba(99, 102, 241, 0.08);
}

/* Glowing Neon Elements */
.glow-text-indigo {
  text-shadow: 0 0 15px rgba(99, 102, 241, 0.5);
}

.glow-text-emerald {
  text-shadow: 0 0 15px rgba(16, 185, 129, 0.5);
}

.glow-text-cyan {
  text-shadow: 0 0 15px rgba(6, 182, 212, 0.5);
}

/* Gradient Borders */
.gradient-border-indigo {
  position: relative;
  border-radius: 1rem;
  background: linear-gradient(to right, #6366f1, #3b82f6);
  padding: 1px;
}

.gradient-border-content {
  background: #0f172a;
  border-radius: 15px;
}

/* Glowing card borders for active selected targets */
.glow-border-indigo {
  box-shadow: 0 0 20px 1px rgba(99, 102, 241, 0.15);
  border-color: rgba(99, 102, 241, 0.35) !important;
}

/* Emerald Pulsing Button Glow */
.pulse-glow-emerald {
  animation: pulseEmerald 2s infinite;
}

@keyframes pulseEmerald {
  0% {
    box-shadow: 0 0 0 0 rgba(16, 185, 129, 0.4);
  }
  70% {
    box-shadow: 0 0 0 10px rgba(16, 185, 129, 0);
  }
  100% {
    box-shadow: 0 0 0 0 rgba(16, 185, 129, 0);
  }
}

/* Gradient Text Highlights */
.gradient-text {
  background: linear-gradient(135deg, #a5b4fc 0%, #6366f1 50%, #22d3ee 100%);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

/* Subtle Card Gradient Overlays */
.card-gradient-overlay {
  background: linear-gradient(180deg, rgba(255, 255, 255, 0.015) 0%, rgba(255, 255, 255, 0) 100%);
}
```

```
/* Blinking Typewriter Cursor */
.cursor-blink {
  animation: cursorBlink 0.8s infinite;
}

@keyframes cursorBlink {
  0%, 100% { opacity: 0; }
  50% { opacity: 1; }
}

/* Radial Glow Background Orb for Hero Landing */
.bg-gradient-orb {
  position: absolute;
  width: 650px;
  height: 650px;
  border-radius: 50%;
  background: radial-gradient(circle, rgba(99, 102, 241, 0.08) 0%, rgba(34, 211, 238, 0.03) 50%, rgba(0,0,0,0) 100%);
  filter: blur(80px);
  z-index: 0;
  top: 40%;
  left: 50%;
  transform: translate(-50%, -50%);
  pointer-events: none;
}

/* Floating animation for Scroll Chevron */
.animate-bounce-slow {
  animation: bounceSlow 2s infinite;
}

@keyframes bounceSlow {
  0%, 100% {
    transform: translateY(0);
  }
  50% {
    transform: translateY(-8px);
  }
}

/* Animations */
.animate-fade-in {
  animation: fadeIn 0.4s ease-out forwards;
}

.animate-slide-up {
  animation: slideUp 0.5s cubic-bezier(0.16, 1, 0.3, 1) forwards;
}

@keyframes fadeIn {
  from {
    opacity: 0;
  }
  to {
    opacity: 1;
  }
}

@keyframes slideUp {
  from {
    opacity: 0;
    transform: translateY(16px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}
```

```
}

/* Table styles */
.styled-table th {
  font-family: &#x27;Outfit&#x27;;, sans-serif;
  letter-spacing: 0.05em;
}

/* Plotly customization overrides */
.js-plotly-plot .plotly .modebar {
  background-color: transparent !important;
}

.js-plotly-plot .plotly .user-select-none {
  cursor: default !important;
}

/* Plotly range slider styling override */
.js-plotly-plot .plotly .slider-container {
  font-size: 10px !important;
}

/* Animations for Dashboard Name Title Sequence */
.letter-zoom-fade {
  animation: letterZoomFade 0.2s cubic-bezier(0.34, 1.56, 0.64, 1) forwards;
  display: inline-block;
}

@keyframes letterZoomFade {
  0% {
    opacity: 0;
    transform: scale(0.3) rotate(-15deg);
    filter: blur(8px);
  }
  50% {
    opacity: 0.8;
    transform: scale(1.15) rotate(5deg);
    filter: blur(2px);
  }
  100% {
    opacity: 1;
    transform: scale(1) rotate(0deg);
    filter: blur(0);
  }
}

.word-glow-in {
  animation: wordGlowIn 1.8s cubic-bezier(0.25, 1, 0.5, 1) forwards;
  display: inline-block;
  background: linear-gradient(135deg, #a5b4fc 0%, #6366f1 50%, #22d3ee 100%);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
}

@keyframes wordGlowIn {
  0% {
    opacity: 0;
    transform: scale(0.85);
    filter: blur(12px);
    text-shadow: 0 0 0px rgba(99, 102, 241, 0);
  }
  12% {
    opacity: 1;
    transform: scale(1.02);
    filter: blur(0);
    text-shadow: 0 0 15px rgba(99, 102, 241, 0.4), 0 0 30px rgba(34, 211, 238, 0.2);
  }
}
```

```
    }
    15%, 85% {
      opacity: 1;
      transform: scale(1);
      filter: blur(0);
      text-shadow: 0 0 30px rgba(99, 102, 241, 0.7), 0 0 60px rgba(34, 211, 238, 0.4), 0 0 90px rgba(99, 102, 241, 0.2);
    }
    95% {
      opacity: 0.2;
      transform: scale(1.03);
      filter: blur(4px);
      text-shadow: 0 0 15px rgba(99, 102, 241, 0.2);
    }
    100% {
      opacity: 0;
      transform: scale(1.05);
      filter: blur(15px);
      text-shadow: 0 0 0px rgba(99, 102, 241, 0);
    }
  }
}

/* Make sure the container holds the height during animation to avoid layout shifts */
.title-container-fixed {
  min-height: 120px;
  display: flex;
  align-items: center;
  justify-content: center;
}
```

■ backend/static/js/components.js

Component Description: React UI component library. Implements navigation headers, stock selectors, uploader zones, metrics diagnostics, and the animated HeroSection.

```
const React = window.React;
const { useState, useEffect } = React;

// SVG Icons
const Icons = {
  Check: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M9 12l5 5 4-4' })
    ),
  },
  Alert: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M12 9v3m0 0h3m-5 4h10' })
    ),
  },
  Download: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M19 13H5m12 0a2 2 0 010 2h3' })
    ),
  },
  Upload: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M12 16l4-4m0 0h3' })
    ),
  },
  Database: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M4 6h16' })
    ),
  },
  Trend: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M12 16l4-4m0 0h3' })
    ),
  },
  Info: ({ className = 'w-4 h-4' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M12 16l4-4m0 0h3' })
    ),
  },
  Loader: ({ className = 'w-5 h-5 animate-spin' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24',
      React.createElement('circle', { className: 'opacity-25', cx: '12', cy: '12',
        React.createElement('path', { className: 'opacity-75', fill: 'currentColor', d: 'M4 4' })
      })
    ),
  },
  Academic: ({ className = 'w-5 h-5' }) => {
    React.createElement('svg', { className, fill: 'none', viewBox: '0 0 24 24', stroke: 'black',
      React.createElement('path', { strokeLinecap: 'round', strokeLinejoin: 'round', d: 'M12 16l4-4m0 0h3' })
    ),
  },
};

// Tooltip helper component
function HelpTooltip({ text }) {
  const [visible, setVisible] = useState(false);
  return React.createElement('span', { className: 'relative inline-block ml-1 cursor-help group text-gray-500',
    React.createElement('span', {
      onMouseEnter: () => setVisible(true),
    })
  ),
};
```

```

    onMouseLeave: () => setVisible(false),
    className: 'inline-flex items-center justify-center w-3.5 h-3.5 text-[10px] font-bold border border-gray-500 rounded',
  }, '<?>'),
  visible && React.createElement('<span>', {
    className: 'absolute bottom-full left-1/2 -translate-x-1/2 mb-2 w-48 p-2 text-xs bg-slate-900 border border-gray-500',
    text
  }, text)
});
}

// 1. Header Component
export function Header() {
  return React.createElement('<header>', { className: 'glass-panel px-6 py-4 flex items-center justify-between' },
    React.createElement('<div>', { className: 'flex items-center gap-3' },
      React.createElement('<div>', { className: 'bg-gradient-to-tr from-indigo-500 to-cyan-400 p-2.5 rounded-l' },
        React.createElement(Icons.Academic, { className: 'w-6 h-6 text-slate-900' })
      ),
      React.createElement('<div>', {},
        React.createElement('<h1>', { className: 'text-xl font-bold tracking-tight text-white flex items-center' },
          '<FAMA AI>',
          React.createElement('<span>', { className: 'text-xs bg-indigo-500/20 text-indigo-400 px-2 py-0.5 rounded' })
        ),
        React.createElement('<p>', { className: 'text-xs text-gray-400' }, '<Fama-French 3-Factor Stock Model>')
      ),
      React.createElement('<div>', { className: 'flex items-center gap-4' },
        React.createElement('<div>', { className: 'hidden sm:flex items-center gap-2 px-3 py-1.5 rounded-full bg-indigo-500/20' },
          React.createElement('<span>', { className: 'w-2 h-2 rounded-full bg-emerald-500 animate-pulse' })
        ),
        React.createElement('<span>', { className: 'text-gray-400 font-medium font-mono' }, '<OLS Eng>')
      )
    ),
  );
}

// 1.5 Quick Start Tutorial Guide (Togglable)
export function QuickGuide() {
  const [open, setOpen] = useState(false);

  return React.createElement('<div>', { className: 'glass-panel p-4 border border-gray-800/80 mb-6 shadow-xl' },
    React.createElement('<div>', { className: 'flex justify-between items-center cursor-pointer select-none' },
      React.createElement('<h4>', { className: 'text-xs font-bold text-indigo-400 uppercase tracking-widest' },
        React.createElement('<span>', { className: 'w-2 h-2 rounded-full bg-indigo-500 animate-pulse' })
      ),
      '<Quick Operating Guide (Click to toggle)>'
    ),
    React.createElement('<span>', { className: 'text-xs text-gray-500 font-bold font-mono' }, open ? '<Close>' : '<Open>')
  ),
  open && React.createElement('<div>', { className: 'grid grid-cols-1 md:grid-cols-3 gap-4 mt-4 border border-gray-900/40' },
    React.createElement('<div>', { className: 'p-3.5 bg-slate-950/40 rounded-xl border border-gray-900' },
      React.createElement('<div>', { className: 'font-semibold text-gray-200 mb-1 flex items-center gap-1.5' },
        React.createElement('<span>', { className: 'w-4.5 h-4.5 bg-indigo-500/20 text-indigo-400 rounded-full' },
          '<Select target stock>'
        ),
        React.createElement('<p>', { className: 'text-gray-400 leading-relaxed font-normal' },
          '<Choose pre-loaded NIFTY 50 data (Infosys, Tata Motors) or click the upload button to parse your custom factor data>'
        )
      ),
      React.createElement('<div>', { className: 'p-3.5 bg-slate-950/40 rounded-xl border border-gray-900' },
        React.createElement('<div>', { className: 'font-semibold text-gray-200 mb-1 flex items-center gap-1.5' },
          React.createElement('<span>', { className: 'w-4.5 h-4.5 bg-indigo-500/20 text-indigo-400 rounded-full' },
            '<Analyze factor tags>'
          ),
          React.createElement('<p>', { className: 'text-gray-400 leading-relaxed font-normal' },
            '<Look at the top horizontal cards to instantly understand if the stock behaves as defensive/aggressive, growth, or value>'
          )
        ),
        React.createElement('<div>', { className: 'p-3.5 bg-slate-950/40 rounded-xl border border-gray-900' },
          React.createElement('<div>', { className: 'font-semibold text-gray-200 mb-1 flex items-center gap-1.5' },
            React.createElement('<span>', { className: 'w-4.5 h-4.5 bg-indigo-500/20 text-indigo-400 rounded-full' },
              '<Analyze factor tags>'
            ),
            React.createElement('<p>', { className: 'text-gray-400 leading-relaxed font-normal' },
              '<Look at the top horizontal cards to instantly understand if the stock behaves as defensive/aggressive, growth, or value>'
            )
          )
        )
      )
    )
  );
}

```

```

        React.createElement(&quot;span&quot;;, { className: &quot;w-4.5 h-4.5 bg-indigo-500/20 text-indigo-400 rounded-full&quot;;
        &quot;Hover over (?) symbols&quot;;
    ),
    React.createElement(&quot;p&quot;;, { className: &quot;text-gray-400 leading-relaxed font-normal&quot;; },
        &quot;Confused by terms like R-squared, t-Stat, or p-Value? Hover your cursor over any question mark for a simple
    )
)
);
}

// 1.8 Hero Landing Page Component
export function HeroSection() {
    const [text, setText] = useState(&quot;&quot;);
    const [wordIndex, setWordIndex] = useState(0);
    const [isDeleting, setIsDeleting] = useState(false);
    const words = [
        &quot;Fama-French Smart Analytics Platform&quot;;,
        &quot;Automated 3-Factor OLS Regression Engine&quot;;,
        &quot;Vibrant Plotly Financial Visualizations&quot;;,
        &quot;MBA + BTech Academic Project Excellence&quot;;
    ];

    const [animIndex, setAnimIndex] = useState(0);
    const [animKey, setAnimKey] = useState(0);

    useEffect(() => {
        const sequence = [
            { text: &quot;F&quot;;, isFull: false, duration: 250 },
            { text: &quot;A&quot;;, isFull: false, duration: 250 },
            { text: &quot;M&quot;;, isFull: false, duration: 250 },
            { text: &quot;A&quot;;, isFull: false, duration: 250 },
            { text: &quot;A&quot;;, isFull: false, duration: 250 },
            { text: &quot;I&quot;;, isFull: false, duration: 250 },
            { text: &quot;FAMA AI&quot;;, isFull: true, duration: 1800 }
        ];

        const currentStep = sequence[animIndex];
        const timer = setTimeout(() => {
            setAnimIndex((prev) => (prev + 1) % sequence.length);
            setAnimKey((prev) => prev + 1);
        }, currentStep.duration);

        return () => clearTimeout(timer);
    }, [animIndex]);

    useEffect(() => {
        let timer;
        const currentWord = words[wordIndex];

        const tick = () => {
            if (!isDeleting) {
                setText(currentWord.substring(0, text.length + 1));
                if (text.length + 1 === currentWord.length) {
                    timer = setTimeout(() => setIsDeleting(true), 2500);
                    return;
                }
            } else {
                setText(currentWord.substring(0, text.length - 1));
                if (text.length - 1 === 0) {
                    setIsDeleting(false);
                    setWordIndex((wordIndex + 1) % words.length);
                    timer = setTimeout(() => {}, 500);
                    return;
                }
            }
        }
    }
}

```

```

    const speed = isDeleting ? 30 : 60;
    timer = setTimeout(tick, speed);
  };

  timer = setTimeout(tick, 100);
  return () => {clearTimeout(timer);
}, [text, isDeleting, wordIndex]);

const handleScrollDown = () => {
  const mainSection = document.getElementById("main-dashboard");
  if (mainSection) {
    mainSection.scrollIntoView({ behavior: "smooth" });
  }
};

return React.createElement("div", { className: "relative min-h-screen flex flex-col justify-center items-center" },
  React.createElement("div", { className: "bg-gradient-orb" }),
  React.createElement("div", { className: "relative z-10 text-center max-w-4xl flex flex-col items-center" },
    React.createElement("div", { className: "bg-indigo-500/10 border border-indigo-500/20 text-indigo-400" },
      "Academic equity research platform"
    ),
    React.createElement("div", { className: "title-container-fixed mb-2" },
      React.createElement("h1", {
        key: animKey,
        className: animIndex === 6
          ? "text-6xl sm:text-8xl font-black tracking-wider word-glow-in"
          : "text-7xl sm:text-9xl font-black letter-zoom-fade text-white drop-shadow-[0_0_20px_rgba(99,102,241,0.35)]"
      },
        animIndex === 6 ? "FAMA AI" : ["F", "A", "M", "A"]
      )
    ),
    React.createElement("div", { className: "h-16 flex items-center justify-center" },
      React.createElement("h2", { className: "text-lg sm:text-2xl font-medium text-gray-300 font-mono tracking-wider" },
        text,
        React.createElement("span", { className: "cursor-blink text-indigo-400 ml-1 font-sans font-bold" })
      ),
      React.createElement("p", { className: "text-xs sm:text-sm text-gray-500 max-w-xl leading-relaxed font-mono" },
        "Analyze NIFTY 50 equities using the Fama-French 3-Factor pricing model. Built using FastAPI, Statsmodels OLS, and Linear Regression"
      ),
      React.createElement("div", { className: "mt-12 animate-slide-up" },
        React.createElement("button", {
          onClick: handleScrollDown,
          className: "bg-gradient-to-r from-indigo-600 to-indigo-700 hover:from-indigo-500 hover:to-indigo-600 text-white"
        },
          "Launch Analytics Dashboard"
        ),
        React.createElement("svg", { className: "w-4 h-4" },
          React.createElement("path", { strokeLinecap: "round", strokeLinejoin: "round", d: "M12 12 12 12" })
        )
      ),
      React.createElement("div", { className: "absolute bottom-8 left-1/2 -translate-x-1/2 cursor-pointer flex" },
        React.createElement("span", { className: "text-[10px] font-bold uppercase tracking-widest font-mono" },
          "ANALYTICS DASHBOARD"
        ),
        React.createElement("svg", { className: "w-5 h-5" },
          React.createElement("path", { strokeLinecap: "round", strokeLinejoin: "round", d: "M12 12 12 12" })
        )
      )
    );
};
}

```

```
// 2. Stock Selection Panel
```

```
export function StockSelector({ stocks, selectedStock, onSelect, onUploadClick }) {
  return React.createElement("div", { className: "glass-panel p-6 shadow-xl border border-gray-800/80 mb-6" },
    React.createElement("h3", { className: "text-sm font-semibold text-gray-400 uppercase tracking-wider mb-2" },
      "Stock Selection Panel"
    )
  );
}
```



```

    React.createElement(Icons.Database, { className: 'w-4 h-4 text-indigo-400' }),
    'Select Analysis Target'
  ),
  React.createElement('div', { className: 'flex flex-col sm:flex-row gap-4 items-stretch sm:items-center' },
    React.createElement('div', { className: 'flex-1 relative' },
      React.createElement('select', {
        value: selectedStock || '',
        onChange: (e) => onSelect(e.target.value),
        className: 'w-full bg-slate-950 text-gray-200 border border-gray-800 rounded-lg py-2.5 px-4 focus:ring-2 focus:ring-indigo-400'
      }),
      React.createElement('option', { value: '', disabled: true }, '-- Select a NIFTY 50 Stock'),
      ...stocks.map((stock) => React.createElement('option', { key: stock.id, value: stock.id }, stock.name)),
      React.createElement('option', { value: 'upload', disabled: true }, 'Upload Custom Factor Dataset (.csv)')
    ),
    React.createElement('button', {
      onClick: onUploadClick,
      className: 'flex items-center justify-center gap-2 bg-gradient-to-r from-indigo-600 to-indigo-700 hover:from-indigo-500 hover:to-indigo-600'
    }),
    React.createElement(Icons.Upload, { className: 'w-4 h-4' }),
    'Upload Custom CSV'
  )
);
}

// 3. CSV Uploader Component
export function UploadComponent({ onUploadSuccess, onClose }) {
  const [dragActive, setDragActive] = useState(false);
  const [error, setError] = useState(null);
  const [loading, setLoading] = useState(false);

  const handleDrag = (e) => {
    e.preventDefault();
    e.stopPropagation();
    if (e.type === 'dragenter' || e.type === 'dragover') {
      setDragActive(true);
    } else if (e.type === 'dragleave') {
      setDragActive(false);
    }
  };

  const processFile = async (file) => {
    if (!file) return;
    if (!file.name.endsWith('.csv')) {
      setError('Only CSV files are accepted.');
```

```

    onUploadSuccess(data);
  } catch (err) {
    setError(err.message);
  } finally {
    setLoading(false);
  }
};

const handleDrop = (e) => {
  e.preventDefault();
  e.stopPropagation();
  setDragActive(false);
  if (e.dataTransfer.files && e.dataTransfer.files[0]) {
    processFile(e.dataTransfer.files[0]);
  }
};

const handleFileChange = (e) => {
  if (e.target.files && e.target.files[0]) {
    processFile(e.target.files[0]);
  }
};

return React.createElement(<div>;, { className: <`fixed inset-0 bg-slate-950/80 backdrop-blur-sm flex items-center justify-center`>; },
  React.createElement(<div>;, { className: <`glass-panel w-full max-w-lg overflow-hidden border border-gray-800`>; },
    React.createElement(<div>;, { className: <`px-6 py-4 bg-slate-900 border-b border-gray-800 flex items-center`>; },
      React.createElement(<h3>;, { className: <`text-base font-bold text-white flex items-center gap-2`>; },
        React.createElement(Icons.Upload, { className: <`w-5 h-5 text-indigo-400`>; }),
        <`Upload Factor Dataset`>;
      ),
      React.createElement(<button>;, {
        onClick: onClose,
        className: <`text-gray-400 hover:text-white transition-colors p-1 hover:bg-slate-800 rounded`>;
      }),
      React.createElement(<svg>;, { className: <`w-5 h-5`>;, fill: <`none`>;, viewBox: <`0 0 24 24`>; },
        React.createElement(<path>;, { strokeLinecap: <`round`>;, strokeLinejoin: <`round`>;, stroke: <`white`>; })
      )
    ),
    React.createElement(<div>;, { className: <`p-6`>; },
      React.createElement(<div>;, {
        onDragEnter: handleDrag,
        onDragOver: handleDrag,
        onDragLeave: handleDrag,
        onDrop: handleDrop,
        className: <`border-2 border-dashed rounded-xl p-8 flex flex-col items-center justify-center text-center cursor-pointer`>;,
        dragActive
          ? <`border-indigo-500 bg-indigo-500/5`>;
          : <`border-gray-800 bg-slate-950/30 hover:border-gray-700 hover:bg-slate-950/50`>;
      }),
      onClick: () => document.getElementById(<`file-upload`>).click()
    ),
    React.createElement(<input>;, {
      id: <`file-upload`>;,
      type: <`file`>;,
      accept: <`.csv`>;,
      className: <`hidden`>;,
      onChange: handleFileChange,
      disabled: loading
    }),
    loading ? React.createElement(<div>;, { className: <`flex flex-col items-center gap-3 py-4`>; },
      React.createElement(Icons.Loader, { className: <`w-10 h-10 text-indigo-400`>; }),
      React.createElement(<p>;, { className: <`text-sm text-indigo-300 font-medium`>; }, <`Running`>
    ) : React.createElement(React.Fragment, {},
      React.createElement(<div>;, { className: <`bg-indigo-500/10 p-4 rounded-full mb-4 text-indigo-400`>; },
        React.createElement(Icons.Upload, { className: <`w-8 h-8`>; })
      )
    )
  )
);

```

```

    ),
    React.createElement(&quot;h4&quot;;, { className: &quot;text-sm font-semibold text-gray-200 mb-1&quot;; }, &quot;or click to bro
    React.createElement(&quot;p&quot;;, { className: &quot;text-xs text-gray-400 mb-4&quot;; }, &quot;or click to bro
    React.createElement(&quot;div&quot;;, { className: &quot;bg-slate-900 border border-gray-800 px-3 py-1.5 rounded
      &quot;Expected Columns: Date, Ri_Rf, Rm_Rf, SMB, HML&quot;;
    )
  )
),
error &amp;&amp; React.createElement(&quot;div&quot;;, { className: &quot;mt-4 p-3.5 bg-rose-500/10 border border-ro
  React.createElement(&quot;div&quot;;, { className: &quot;text-rose-400 mt-0.5&quot;; }, React.createElement(Icons.A
  React.createElement(&quot;div&quot;;, { },
    React.createElement(&quot;p&quot;;, { className: &quot;font-semibold mb-0.5&quot;; }, &quot;Upload Failed&quot;;),
    React.createElement(&quot;p&quot;;, { className: &quot;leading-relaxed&quot;; }, error)
  )
),
React.createElement(&quot;div&quot;;, { className: &quot;mt-6 flex justify-end gap-3&quot;; },
  React.createElement(&quot;button&quot;;, {
    onClick: onClose,
    className: &quot;px-4 py-2 border border-gray-800 hover:bg-slate-900 hover:border-gray-700 text-gray-300 font-s
  }, &quot;Cancel&quot;;)
)
)
);
}

// 4. Beginner-Friendly Diagnostics (Metrics Card Strip)
export function ModelDiagnostics({ results }) {
  if (!results) return null;
  const { r_squared, factors } = results;

  const mktBeta = factors.Market.coefficient;
  const smbBeta = factors.SMB.coefficient;
  const hmlBeta = factors.HML.coefficient;

  // Derive simple badges
  const getMktBadge = (val) => {
    if (val > 1.1) return { text: &quot;Aggressive Risk&quot;;, style: &quot;text-rose-400 bg-rose-500/10 border-rose-500
    if (val < 0.9) return { text: &quot;Defensive / Low Risk&quot;;, style: &quot;text-emerald-400 bg-emerald-500/10 bord
    return { text: &quot;Market Risk Match&quot;;, style: &quot;text-indigo-400 bg-indigo-500/10 border-indigo-500/20&quot;;
  };

  const getCapBadge = (val) => {
    if (val > 0.15) return { text: &quot;Small-Cap Stock&quot;;, style: &quot;text-orange-400 bg-orange-500/10 border-ora
    if (val < -0.15) return { text: &quot;Large-Cap Giant&quot;;, style: &quot;text-blue-400 bg-blue-500/10 border-blue-5
    return { text: &quot;Size Neutral&quot;;, style: &quot;text-gray-400 bg-gray-500/10 border-gray-800&quot;; };
  };

  const getStyleBadge = (val) => {
    if (val > 0.15) return { text: &quot;Value stock (Undervalued)&quot;;, style: &quot;text-amber-400 bg-amber-500/10 bo
    if (val < -0.15) return { text: &quot;Growth stock (Premium)&quot;;, style: &quot;text-fuchsia-400 bg-fuchsia-500/10
    return { text: &quot;Blend Stock Style&quot;;, style: &quot;text-gray-400 bg-gray-500/10 border-gray-800&quot;; };
  };

  const getReliability = (val) => {
    if (val > 0.85) return { text: &quot;Highly Reliable&quot;;, style: &quot;text-emerald-400 bg-emerald-500/10 border-e
    if (val >= 0.50) return { text: &quot;Moderate Fit&quot;;, style: &quot;text-amber-400 bg-amber-500/10 border-amber-5
    return { text: &quot;Low Explanatory Power&quot;;, style: &quot;text-rose-400 bg-rose-500/10 border-rose-500/20&quot;; };
  };

  const mkt = getMktBadge(mktBeta);
  const cap = getCapBadge(smbBeta);
  const style = getStyleBadge(hmlBeta);
  const rel = getReliability(r_squared);

  return React.createElement(&quot;div&quot;;, { className: &quot;grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4 mb-6

```

```

// Card 1: Market Risk
React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-4 border border-gray-800/80 flex flex-col justify
  React.createElement(&quot;div&quot;;, { },
    React.createElement(&quot;div&quot;;, { className: &quot;flex justify-between items-center mb-1&quot;; },
      React.createElement(&quot;span&quot;;, { className: &quot;text-xs font-semibold text-gray-400 uppercase tracking-w
      React.createElement(HelpTooltip, { text: &quot;Measures sensitivity to overall market index (NIFTY 50). Beta &gt;
    ),
    React.createElement(&quot;h4&quot;;, { className: &quot;text-xl font-bold text-white font-mono&quot;; }, mktBeta.toFi
  ),
  React.createElement(&quot;div&quot;;, { className: &quot;mt-2&quot;; },
    React.createElement(&quot;span&quot;;, { className: `text-[10px] font-bold px-2 py-0.5 rounded border ${mkt.style}`
  )
),
// Card 2: Company Size
React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-4 border border-gray-800/80 flex flex-col justify
  React.createElement(&quot;div&quot;;, { },
    React.createElement(&quot;div&quot;;, { className: &quot;flex justify-between items-center mb-1&quot;; },
      React.createElement(&quot;span&quot;;, { className: &quot;text-xs font-semibold text-gray-400 uppercase tracking-w
      React.createElement(HelpTooltip, { text: &quot;Positive SMB means the stock responds like a smaller-cap stock. Ne
    ),
    React.createElement(&quot;h4&quot;;, { className: &quot;text-xl font-bold text-white font-mono&quot;; }, smbBeta.toFi
  ),
  React.createElement(&quot;div&quot;;, { className: &quot;mt-2&quot;; },
    React.createElement(&quot;span&quot;;, { className: `text-[10px] font-bold px-2 py-0.5 rounded border ${cap.style}`
  )
),
// Card 3: Stock Style
React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-4 border border-gray-800/80 flex flex-col justify
  React.createElement(&quot;div&quot;;, { },
    React.createElement(&quot;div&quot;;, { className: &quot;flex justify-between items-center mb-1&quot;; },
      React.createElement(&quot;span&quot;;, { className: &quot;text-xs font-semibold text-gray-400 uppercase tracking-w
      React.createElement(HelpTooltip, { text: &quot;Positive HML means it is a value stock (cheap relative to book val
    ),
    React.createElement(&quot;h4&quot;;, { className: &quot;text-xl font-bold text-white font-mono&quot;; }, hmlBeta.toFi
  ),
  React.createElement(&quot;div&quot;;, { className: &quot;mt-2&quot;; },
    React.createElement(&quot;span&quot;;, { className: `text-[10px] font-bold px-2 py-0.5 rounded border ${style.style}
  )
),
// Card 4: Model Accuracy
React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-4 border border-gray-800/80 flex flex-col justify
  React.createElement(&quot;div&quot;;, { },
    React.createElement(&quot;div&quot;;, { className: &quot;flex justify-between items-center mb-1&quot;; },
      React.createElement(&quot;span&quot;;, { className: &quot;text-xs font-semibold text-gray-400 uppercase tracking-w
      React.createElement(HelpTooltip, { text: &quot;R-squared shows what percentage of the stock's return movemen
    ),
    React.createElement(&quot;h4&quot;;, { className: &quot;text-xl font-bold text-white font-mono&quot;; }, `${(r_square
  ),
  React.createElement(&quot;div&quot;;, { className: &quot;mt-2&quot;; },
    React.createElement(&quot;span&quot;;, { className: `text-[10px] font-bold px-2 py-0.5 rounded border ${rel.style}`
  )
),
);
}

// 5. OLS Coefficients Table (with Tooltips)
export function ResultsTable({ results }) {
  if (!results) return null;
  const { factors } = results;

  const getPValueStyle = (val) => {
    if (val < 0.05) return &quot;text-emerald-400 font-semibold&quot;;
    return &quot;text-gray-400&quot;;
  };

  const getFactorLabel = (factorKey) => {

```

```

return {
  &quot;Alpha&quot;;: &quot;Alpha (Intercept)&quot;;,
  &quot;Market&quot;;: &quot;Market Factor (Rm - Rf)&quot;;,
  &quot;SMB&quot;;: &quot;Size Factor (SMB)&quot;;,
  &quot;HML&quot;;: &quot;Style Factor (HML)&quot;;
}[factorKey] || factorKey;
};

return React.createElement(&quot;div&quot;;, { className: &quot;glass-panel overflow-hidden border border-gray-800/80 shadow-2xl&quot;; },
  React.createElement(&quot;div&quot;;, { className: &quot;px-6 py-4 bg-slate-900/50 border-b border-gray-800 flex items-center&quot;; },
    React.createElement(&quot;h3&quot;;, { className: &quot;text-sm font-semibold text-gray-200 tracking-wide uppercase&quot;; },
      React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-3 bg-indigo-500 rounded-full&quot;; }),
      &quot;Statistical Regression Results (OLS)&quot;;
    ),
  ),
  React.createElement(&quot;div&quot;;, { className: &quot;overflow-x-auto text-xs sm:text-sm&quot;; },
    React.createElement(&quot;table&quot;;, { className: &quot;w-full text-left border-collapse styled-table&quot;; },
      React.createElement(&quot;thead&quot;;, { },
        React.createElement(&quot;tr&quot;;, { className: &quot;border-b border-gray-800 bg-slate-950/40 text-gray-400&quot;; },
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5&quot;; },
            &quot;Factor Name&quot;;,
            React.createElement(HelpTooltip, { text: &quot;The systemic market factor tested in the model.&quot;; })
          ),
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5 text-right&quot;; },
            &quot;Beta Coefficient&quot;;,
            React.createElement(HelpTooltip, { text: &quot;The slope or loading coefficient. A positive coefficient indicates a positive relationship with the market.&quot;; })
          ),
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5 text-right&quot;; },
            &quot;t-Stat&quot;;,
            React.createElement(HelpTooltip, { text: &quot;The test statistic. Values below -2 or above +2 indicate that the factor is statistically significant.&quot;; })
          ),
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5 text-right&quot;; },
            &quot;p-Value&quot;;,
            React.createElement(HelpTooltip, { text: &quot;The statistical probability of finding this result by chance.&quot;; })
          ),
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5 text-center&quot;; },
            &quot;95% Confidence Interval&quot;;,
            React.createElement(HelpTooltip, { text: &quot;The range in which the true factor loading lies with 95% mathematical probability.&quot;; })
          ),
          React.createElement(&quot;th&quot;;, { className: &quot;py-3.5 px-5 text-center&quot;; },
            &quot;Significant?&quot;;,
            React.createElement(HelpTooltip, { text: &quot;Reliable? Yes if p-value is less than 0.05, meaning this factor is statistically significant.&quot;; })
          )
        ),
      ),
    React.createElement(&quot;tbody&quot;;, { className: &quot;divide-y divide-gray-800/50&quot;; },
      Object.keys(factors).map((key) => {
        const factor = factors[key];
        return React.createElement(&quot;tr&quot;;, { key, className: &quot;hover:bg-slate-900/25 transition-colors&quot;; },
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 font-semibold text-gray-300&quot;; }, getFactorName(factor)),
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 text-right font-mono text-gray-100 font-size-0.9&quot;; }, factor.beta),
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 text-right font-mono text-gray-300&quot;; }, factor.tStat),
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 text-right font-mono ${getPValueStyle(factor.pValue)} text-gray-300&quot;; }, factor.pValue),
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 text-center font-mono text-gray-400&quot;; }, factor.confInt),
          React.createElement(&quot;td&quot;;, { className: &quot;py-3.5 px-5 text-center&quot;; },
            factor.significant
              ? React.createElement(&quot;span&quot;;, { className: &quot;inline-flex items-center gap-1.5 px-2 py-0.5 rounded&quot;; },
                  React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-1.5 rounded-full bg-emerald-400&quot;; }),
                  &quot;Reliable&quot;;
                )
              : React.createElement(&quot;span&quot;;, { className: &quot;inline-flex items-center gap-1.5 px-2 py-0.5 rounded&quot;; },
                  React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-1.5 rounded-full bg-gray-500&quot;; }),
                  &quot;Unreliable&quot;;
                )
            )
          )
        )
      })
    )
  );
};

```

```

    })
  )
)
);
}

// 6. Interpretation Panel Component
export function InterpretationPanel({ results }) {
  if (!results) return null;
  const { interpretations, factors } = results;

  const getStyleTag = (key) => {
    if (key === '"market"') {
      const val = factors.Market.coefficient;
      if (val > 1.1) return { text: '"Aggressive"', color: '"bg-rose-500/10 text-rose-400 border-rose-500/20' };
      if (val < 0.9) return { text: '"Defensive"', color: '"bg-emerald-500/10 text-emerald-400 border-emerald-500/20' };
      return { text: '"Market Core"', color: '"bg-indigo-500/10 text-indigo-400 border-indigo-500/20' };
    }
    if (key === '"smb"') {
      const val = factors.SMB.coefficient;
      if (val > 0.15) return { text: '"Small-Cap Profile"', color: '"bg-orange-500/10 text-orange-400 border-orange-500/20' };
      if (val < -0.15) return { text: '"Mega-Cap Giant"', color: '"bg-blue-500/10 text-blue-400 border-blue-500/20' };
      return { text: '"Size Neutral"', color: '"bg-gray-500/10 text-gray-400 border-gray-800' };
    }
    if (key === '"hml"') {
      const val = factors.HML.coefficient;
      if (val > 0.15) return { text: '"Value Style"', color: '"bg-amber-500/10 text-amber-400 border-amber-500/20' };
      if (val < -0.15) return { text: '"Growth Style"', color: '"bg-fuchsia-500/10 text-fuchsia-400 border-fuchsia-500/20' };
      return { text: '"Blend Style"', color: '"bg-gray-500/10 text-gray-400 border-gray-800' };
    }
    return { text: '"Insight"', color: '"bg-gray-500/10 text-gray-400 border-gray-800' };
  };

  const getTitle = (key) => {
    return {
      '"market"': '"Market Sensitivity (Beta)"',
      '"smb"': '"Company Size Exposure (SMB)"',
      '"hml"': '"Investment Style (HML)"',
      '"alpha"': '"Abnormal Outperformance (Alpha)"',
      '"r_squared"': '"Model Reliability (R²)"'
    }[key] || key;
  };

  return React.createElement('"div"', { className: '"glass-panel p-6 border border-gray-800/80 shadow-xl mb-6' },
    React.createElement('"div"', { className: '"flex items-center gap-2 border-b border-gray-800 pb-4 mb-4' },
      React.createElement('"div"', { className: '"text-indigo-400 bg-indigo-500/10 p-2 rounded-lg animate-pulse' },
        React.createElement(Icons.Trend, {})),
      React.createElement('"div"', {},
        React.createElement('"h3"', { className: '"text-sm font-bold text-white uppercase tracking-wider' },
          React.createElement('"p"', { className: '"text-xs text-gray-400' }, '"Simplified translation of factors' ))),
      React.createElement('"div"', { className: '"space-y-4' },
        Object.keys(interpretations).map((key) => {
          const tag = getStyleTag(key);
          return React.createElement('"div"', { key, className: '"p-4 bg-slate-950/40 border border-gray-900 rounded' },
            React.createElement('"div"', { className: '"flex flex-wrap items-center justify-between gap-2 mb-2' },
              React.createElement('"span"', { className: '"text-xs font-semibold text-gray-300 tracking-wide' },
                React.createElement('"span"', { className: `text-[10px] font-bold px-2 py-0.5 rounded border ${tag.color}` }, tag.text)),
              React.createElement('"p"', { className: '"text-xs sm:text-sm text-gray-400 leading-relaxed font-normal' })),
            React.createElement('"p"', { className: '"text-xs sm:text-sm text-gray-400 leading-relaxed font-normal' })),
          React.createElement('"p"', { className: '"text-xs sm:text-sm text-gray-400 leading-relaxed font-normal' })),
        ))),
    React.createElement('"p"', { className: '"text-xs sm:text-sm text-gray-400 leading-relaxed font-normal' })),
  );
}

```

```

    );
  }

// 7. Report Export Component
export function ExportReportButton({ stockName, results }) {
  const [exporting, setExporting] = useState(false);

  const handleExport = async () => {
    if (!results || exporting) return;
    setExporting(true);

    try {
      const response = await fetch("/api/reports/export", {
        method: "POST",
        headers: {
          "Content-Type": "application/json",
        },
        body: JSON.stringify({
          stock_name: stockName,
          results: results,
        }),
      });

      if (!response.ok) {
        throw new Error("Failed to download PDF report");
      }

      const blob = await response.blob();
      const url = window.URL.createObjectURL(blob);
      const a = document.createElement("a");
      a.href = url;
      a.download = `Fama_French_Report_${stockName.replace(/\s+/g, "_")}.pdf`;
      document.body.appendChild(a);
      a.click();
      window.URL.revokeObjectURL(url);
      document.body.removeChild(a);
    } catch (err) {
      alert(`Export Failed: ${err.message}`);
    } finally {
      setExporting(false);
    }
  };

  return React.createElement("button", {
    onClick: handleExport,
    disabled: exporting,
    className: "flex items-center justify-center gap-2 bg-gradient-to-r from-emerald-600 to-emerald-700 hover:from-emerald-500 hover:to-emerald-600 text-white",
  },
    exporting ? React.createElement(React.Fragment, {},
      React.createElement(Icons.Loader, { className: "w-4 h-4 text-white" }),
      " Creating PDF..."
    ) : React.createElement(React.Fragment, {},
      React.createElement(Icons.Download, { className: "w-4 h-4" }),
      " Download PDF Report"
    )
  );
}

```

□ backend/static/js/app.js

Component Description: Main React client state manager. Coordinates API fetching, chart configuration binds, and overall application bootstrap routing.

```
// =====
// IMPORTS (CDN STYLE SAFE)
// =====
import {
  Header,
  QuickGuide,
  HeroSection,
  StockSelector,
  UploadComponent,
  ResultsTable,
  ModelDiagnostics,
  InterpretationPanel,
  ExportReportButton
} from &quot;./components.js&quot;;

const React = window.React;
const ReactDOM = window.ReactDOM;
const Plotly = window.Plotly;
const {
  useState,
  useEffect,
  useRef
} = React;

// =====
// CHART 1 - EXCESS RETURN TREND
// =====
function ExcessReturnTrendChart({ rawData }) {
  const chartRef = useRef(null);

  useEffect(() => {
    if (!chartRef.current || !rawData || rawData.length === 0) return;

    const dates = rawData.map(d => d.Date);
    const riRf = rawData.map(d => d.Ri_Rf);
    const rmRf = rawData.map(d => d.Rm_Rf);

    const traces = [
      {
        x: dates,
        y: riRf,
        name: &quot;Stock Excess Return&quot;;,
        type: &quot;scatter&quot;;,
        mode: &quot;lines+markers&quot;;,
        line: { color: &quot;#6366f1&quot;;, width: 2.5 },
        marker: { size: 5, color: &quot;#818cf8&quot;; },
        fill: &quot;tozero&quot;;,
        fillcolor: &quot;rgba(99, 102, 241, 0.05)&quot;;
      },
      {
        x: dates,
        y: rmRf,
        name: &quot;Market Excess Return&quot;;,
        type: &quot;scatter&quot;;,
        mode: &quot;lines&quot;;,
        line: { color: &quot;#06b6d4&quot;;, width: 1.5, dash: &quot;dash&quot;; }
      }
    ];
  });
}
```



```

Plotly.newPlot(chartRef.current, traces, {
  paper_bgcolor: '"transparent"',
  plot_bgcolor: '"transparent"',
  margin: { t: 20, l: 40, r: 20, b: 40 },
  font: { color: '"#94a3b8"', family: '"Inter, sans-serif"' },
  hovermode: '"x unified"',
  xaxis: {
    gridcolor: '"#1e293b"',
    tickcolor: '"#334155"',
    zeroline: false,
    ranglider: { visible: true },
    type: '"date"',
  },
  yaxis: {
    gridcolor: '"#1e293b"',
    tickcolor: '"#334155"',
    zerolinecolor: '"#475569"',
  },
  legend: {
    orientation: '"h"',
    yanchor: '"bottom"',
    y: 1.02,
    xanchor: '"right"',
    x: 1
  }
}, { responsive: true, displayModeBar: false });
}, [rawData]);

return React.createElement('"div"', { ref: chartRef, className: '"w-full h-80"' });
}

```

```

// =====
// CHART 2 - FACTOR EXPOSURE
// =====

```

```

function FactorExposureChart({ factors }) {
  const chartRef = useRef(null);

```

```

  useEffect(() => {
    if (!chartRef.current || !factors) return;

```

```

    const values = [
      factors.Market.coefficient,
      factors.SMB.coefficient,
      factors.HML.coefficient
    ];

```

```

    Plotly.newPlot(chartRef.current, [{
      x: ['"Market Sensitivity (Rm-Rf)"', '"Size Sensitivity (SMB)"', '"Style Sensitivity (HML)"'],
      y: values,
      type: '"bar"',
      marker: {
        color: ['"#6366f1"', '"#06b6d4"', '"#10b981"'],
        line: { color: '"rgba(255,255,255,0.05)"', width: 1 }
      },
      text: values.map(v => v.toFixed(3)),
      textposition: '"outside"',
      textfont: { color: '"#f3f4f6"', family: '"Fira Code, monospace"' }
    }], {

```

```

      paper_bgcolor: '"transparent"',
      plot_bgcolor: '"transparent"',
      font: { color: '"#94a3b8"', family: '"Inter, sans-serif"' },
      margin: { t: 25, l: 40, r: 20, b: 30 },
      xaxis: { gridcolor: '"transparent"' },
      yaxis: { gridcolor: '"#1e293b"', zerolinecolor: '"#475569"' }
    }, { responsive: true, displayModeBar: false });

```

```

    }, [factors]);

    return React.createElement(&quot;div&quot;;, { ref: chartRef, className: &quot;w-full h-80&quot;; });
}

// =====
// CHART 3 - P VALUE CHART
// =====
function PValueComparisonChart({ factors }) {
    const chartRef = useRef(null);

    useEffect(() => {
        if (!chartRef.current || !factors) return;

        const pvals = [
            factors.Market.p_value,
            factors.SMB.p_value,
            factors.HML.p_value
        ];

        const colors = pvals.map(p => p < 0.05 ? &quot;#10b981&quot;; : &quot;#64748b&quot;);

        Plotly.newPlot(chartRef.current, [{
            x: [&quot;Market (Rm-Rf)&quot;;, &quot;Size (SMB)&quot;;, &quot;Style (HML)&quot;;],
            y: pvals,
            type: &quot;bar&quot;;,
            marker: { color: colors },
            text: pvals.map(p => p.toFixed(4)),
            textposition: &quot;outside&quot;;,
            textfont: { color: &quot;#f3f4f6&quot;;, family: &quot;Fira Code, monospace&quot;; }
        ]], {
            paper_bgcolor: &quot;transparent&quot;;,
            plot_bgcolor: &quot;transparent&quot;;,
            font: { color: &quot;#94a3b8&quot;;, family: &quot;Inter, sans-serif&quot;; },
            margin: { t: 30, l: 40, r: 20, b: 30 },
            xaxis: { gridcolor: &quot;transparent&quot;; },
            yaxis: {
                gridcolor: &quot;#1e293b&quot;;,
                zerolinecolor: &quot;#475569&quot;;,
                range: [0, Math.max(0.1, ...pvals) * 1.15]
            },
            shapes: [{
                type: &quot;line&quot;;,
                x0: 0,
                x1: 1,
                y0: 0.05,
                y1: 0.05,
                xref: &quot;paper&quot;;,
                yref: &quot;y&quot;;,
                line: { color: &quot;#ef4444&quot;;, width: 1.5, dash: &quot;dash&quot;; }
            }]
        }, { responsive: true, displayModeBar: false });

    }, [factors]);

    return React.createElement(&quot;div&quot;;, { ref: chartRef, className: &quot;w-full h-80&quot;; });
}

// =====
// MAIN APP
// =====
function App() {
    const [stocks, setStocks] = useState([]);

```

```

const [selectedStock, setSelectedStock] = useState('');
const [currentAnalysis, setCurrentAnalysis] = useState(null);
const [loading, setLoading] = useState(false);
const [error, setError] = useState(null);
const [showUpload, setShowUpload] = useState(false);

// Load stocks
useEffect(() => {
  fetch('/api/stocks')
    .then(res => res.json())
    .then(data => {
      setStocks(data);
      if (data.length > 0) {
        handleStockChange(data[0].id);
      }
    })
    .catch(err => setError(err.message));
}, []);

const handleStockChange = async (id) => {
  if (id === 'upload') {
    setShowUpload(true);
    return;
  }

  setSelectedStock(id);
  setLoading(true);
  setError(null);

  try {
    const res = await fetch(`/api/fama-analysis/${id}`);
    const data = await res.json();
    setCurrentAnalysis(data);
  } catch (err) {
    setError(err.message);
  }

  setLoading(false);
};

const handleUploadSuccess = (data) => {
  setCurrentAnalysis(data);
  setSelectedStock('');
  setShowUpload(false);
};

return React.createElement('div', { className: 'bg-slate-950 min-h-screen text-white },
  React.createElement(HeroSection),
  React.createElement('div', { id: 'main-dashboard', className: 'min-h-screen px-4 md:px-8 py-6 },
    React.createElement(Header),
    React.createElement(QuickGuide),

    error && React.createElement('div', { className: 'bg-red-500/20 p-3 rounded mb-4 text-sm border },

    React.createElement(StockSelector, {
      stocks,
      selectedStock,
      onSelect: handleStockChange,
      onUploadClick: () => setShowUpload(true)
    }),
    loading ? React.createElement('div', { className: 'flex-1 flex flex-col items-center justify-center py-6

```

```

    React.createElement(&quot;svg&quot;;, { className: &quot;w-10 h-10 text-indigo-500 animate-spin&quot;;, fill: &quot;n
    React.createElement(&quot;circle&quot;;, { className: &quot;opacity-25&quot;;, cx: &quot;12&quot;;, cy: &quot;12&quot;;
    React.createElement(&quot;path&quot;;, { className: &quot;opacity-75&quot;;, fill: &quot;currentColor&quot;;, d: &qu
  ),
  React.createElement(&quot;p&quot;;, { className: &quot;text-gray-400 font-semibold text-sm&quot;; }, &quot;Running OL
) : currentAnalysis &amp;&amp; React.createElement(&quot;div&quot;;, { className: &quot;space-y-6 animate-fade-in&quot;;

// Stock title banner card
React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-5 border border-gray-800 flex flex-col sm:fl
  React.createElement(&quot;div&quot;;, {
    React.createElement(&quot;span&quot;;, { className: &quot;text-[10px] font-bold text-indigo-400 uppercase tracki
    React.createElement(&quot;h2&quot;;, { className: &quot;text-2xl font-extrabold text-white mt-1.5 tracking-tight
  ),
  React.createElement(ExportReportButton, {
    stockName: currentAnalysis.stock_name,
    results: currentAnalysis.results
  })
),

// Beginner-friendly Card diagnostics
React.createElement(ModelDiagnostics, { results: currentAnalysis.results }),

// Grid Row: Results Table &amp; Explanations side-by-side
React.createElement(&quot;div&quot;;, { className: &quot;grid grid-cols-1 lg:grid-cols-3 gap-6&quot;; },
  React.createElement(&quot;div&quot;;, { className: &quot;lg:col-span-2 flex flex-col justify-between&quot;; },
    React.createElement(ResultsTable, { results: currentAnalysis.results }),

    React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-6 border border-gray-800/80 shadow-xl fle
      React.createElement(&quot;h3&quot;;, { className: &quot;text-sm font-semibold text-gray-200 uppercase tracking
      React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-3 bg-indigo-500 rounded-full&quot;; }),
      &quot; Stock Excess Return Trend (Ri - Rf)&quot;;
    ),
    React.createElement(ExcessReturnTrendChart, { rawData: currentAnalysis.data })
  ),
  React.createElement(&quot;div&quot;;, {
    React.createElement(InterpretationPanel, { results: currentAnalysis.results })
  })
),

// Grid Row: Bar exposure charts side-by-side (No blank columns!)
React.createElement(&quot;div&quot;;, { className: &quot;grid grid-cols-1 md:grid-cols-2 gap-6&quot;; },
  React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-6 border border-gray-800/80 shadow-xl&quot;;
    React.createElement(&quot;h3&quot;;, { className: &quot;text-sm font-semibold text-gray-200 uppercase tracking-w
    React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-3 bg-indigo-500 rounded-full&quot;; }),
    &quot; Factor Exposure Strength (Beta Loading)&quot;;
  ),
  React.createElement(FactorExposureChart, { factors: currentAnalysis.results.factors })
),
  React.createElement(&quot;div&quot;;, { className: &quot;glass-panel p-6 border border-gray-800/80 shadow-xl&quot;;
    React.createElement(&quot;h3&quot;;, { className: &quot;text-sm font-semibold text-gray-200 uppercase tracking-w
    React.createElement(&quot;span&quot;;, { className: &quot;w-1.5 h-3 bg-indigo-500 rounded-full&quot;; }),
    &quot; Factor Significance (p-Value)&quot;;
  ),
  React.createElement(&quot;p&quot;;, { className: &quot;text-[11px] text-gray-500 mb-4&quot;; }, &quot;Values below
  React.createElement(PValueComparisonChart, { factors: currentAnalysis.results.factors })
)
),

React.createElement(&quot;footer&quot;;, { className: &quot;mt-12 py-6 border-t border-gray-900 text-center text-xs te
  React.createElement(&quot;p&quot;;, { }, &quot;© 2026 FAMA AI Platform. Built for MBA + BTech Academic Research &amp;
),

showUpload &amp;&amp; React.createElement(UploadComponent, {
  onUploadSuccess: handleUploadSuccess,

```

```
        onClose: () => setShowUpload(false)
      })
    )
  };
}

// =====
// SAFE RENDER (FIXED)
// =====
const rootEl = document.getElementById("root");

if (rootEl) {
  const root = ReactDOM.createRoot(rootEl);
  root.render(React.createElement(App));
}
```